

Sep. 21st 2026 Week 3 Paper-to-Peer

Beyond Geometry: Comparing the Temporal Structure of Computation in Neural Circuits with Dynamical Similarity Analysis

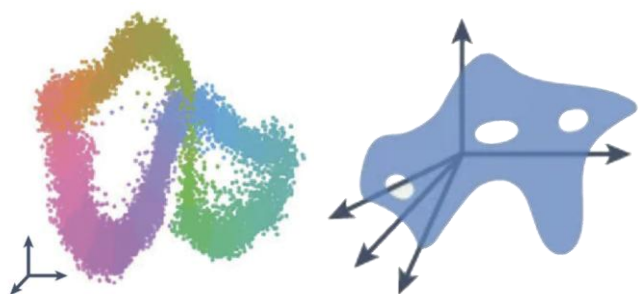
(Ostrow et al., 2023)

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Motivation

How can we tell whether two dynamical systems utilize the *same internal processes* for a particular computation?

Conventional similarity measures



Topology / Geometry / Dynamics

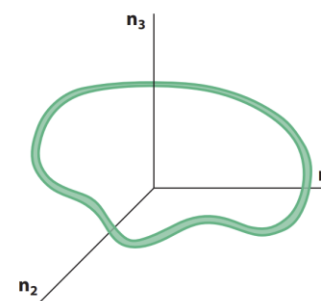
They mostly compared(R^2) the *geometry* of states in the latent space* of a neural network.

- **Fixed point manifolds**

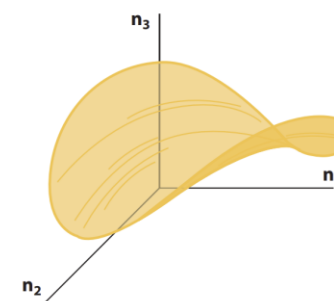
Fixed points are points in state space where repeated iteration of f does not change the state. Perturbations away from the fixed points collapse back to the fixed points.

$$\frac{dx}{dt} = f(x(t)) \Rightarrow f(x^*(t)) = 0$$

b 1D manifold ring



c 2D manifold

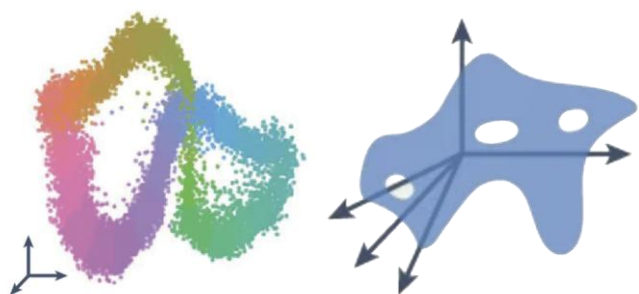


* **Latent space:** low-dimensional mathematical space where high-dimensional data is compressed to capture its core hidden features

Motivation

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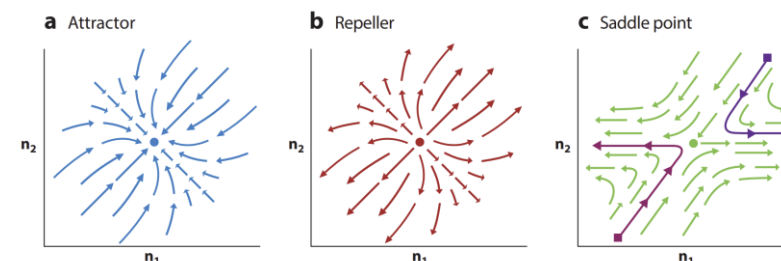
They mostly compared(R^2) the *geometry* of states in the latent space* of a neural network.

But...

- Geometries can differ due to sampling differences or other fundamental causes even though the dynamics are *topologically equivalent*
→ False negative



- System may exhibit *distinct dynamics* over *similar geometries* in the state space
→ False positive



Computations are implemented at the level of *dynamics*!
→ **Dynamical Similarity Analysis (DSA)**

Some Math

How can we define “being the same”?

Step 1

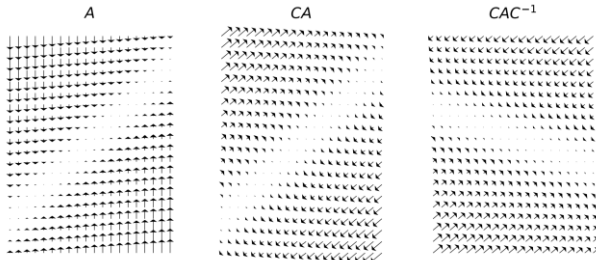
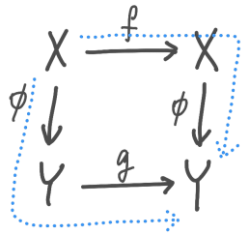
Conjugacy

In a group G , $a, b \in G$ are conjugate if $\exists g \in G$ s.t. $a = gb g^{-1}$

Topological Conjugacy

Given two dynamical systems $f: X \rightarrow X$ and $g: Y \rightarrow Y$ with invertible mapping $\phi: X \rightarrow Y$, conjugacy is defined as

$$g \circ \phi = \phi \circ f$$



→ Coordinate conversion from a system to another.
If conjugate, two systems have the **same qualitative behavior**.

Step 2

Koopman Operator

For a discrete-time dynamical system $x_{t+1} = F(x_t)$, the Koopman operator K is defined by

$$K \circ g = g \circ F$$

- g is an “observable”:
a function that maps the state to a scalar $g: X \rightarrow \mathbb{R}$

$$g_1(x) = x, g_2(x) = \sin(x) \\ \Rightarrow g(x) = [g_1(x) \ g_2(x) \ \dots \ g_m(x)]^T$$

- K is a linear operator:

$$K(\alpha g + \beta h) = (\alpha g + \beta h) \circ F = \alpha(g \circ F) + \beta(h \circ F)$$

It simplifies the problem into $g_{t+1} = K g_t$ ($g_t = g(x_t)$).

→ Enables modeling of nonlinear dynamics into **linear systems**,
which are more tractable & interpretable

Some Math

How can we define “being the same”?

Step 3

Dynamic Mode Decomposition (DMD)

DMD identifies a linear transition operator for the dynamics:

$$X(t + \Delta t) = AX(t)$$

$$A_{DMD} = X_{t+\Delta t} X_t^\dagger = X_{t+\Delta t} V \Sigma^\dagger U^T$$

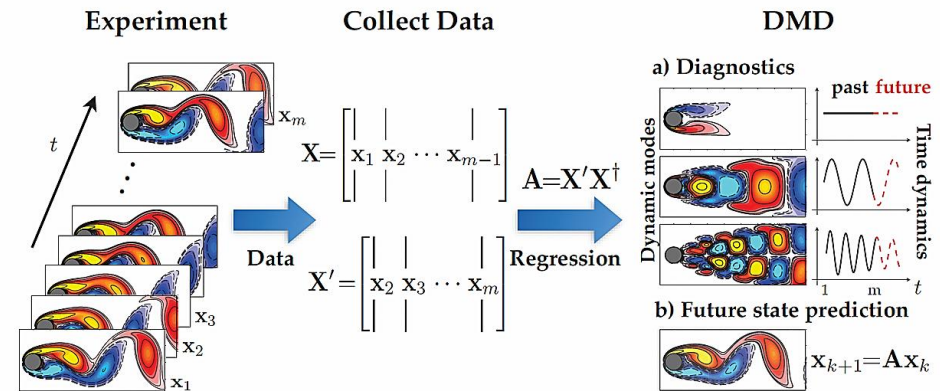
$$\Rightarrow x_n = \sum_{i=1}^M c_i \lambda_i^k v_i \quad (A v_i = \mu_i v_i \quad \forall i = 1, \dots, M)$$

- Koopman operator is theoretically infinite-dimension:

$$g_{t+1} = K g_t \stackrel{?}{\Rightarrow} K = g_{t+1} g_t^{-1}$$

$$G = \{g: \mathbb{R} \rightarrow \mathbb{R}\} \quad (\text{e.g., } 1, x, x^2, x^4, \dots)$$

→ Can use to set a **finite-dimensional** embedding that approximate the Koopman operator



Hankel Matrix

$$X = [-x(t) \quad -x(t - \tau) \quad \dots \quad -x(t - p\tau) \quad -]^T$$

$$\text{e.g., } X = \begin{bmatrix} x(t) & x(t + \tau) & x(t + 2\tau) & \dots \\ x(t - \tau) & x(t) & x(t + \tau) & \dots \\ x(t - 2\tau) & x(t - \tau) & x(t) & \dots \end{bmatrix}$$

- X is closed under the operation of A : $X \in G \Rightarrow AX \in G$
- Improves estimation capabilities over partially-observed systems

Some Math

How can we define “being the same”?

- **Comparing Koopman operator reflects dynamical similarity.**

Theorem 3 (Spectral equivalence of topologically conjugate systems Budišić et al. [2012]). *Let $S : M \rightarrow M$ and $T : N \rightarrow N$ be maps that define two topologically conjugate dynamical systems over different manifolds M and N . Let there exist a homeomorphism $h : N \rightarrow M$ such that $S \circ h = h \circ T$. If ϕ is an eigenfunction of the Koopman operator of S , U_s (i.e. $U_s \phi = \lambda \phi$), then $\phi \circ h$ is an eigenfunction of the Koopman operator of T , U_t also with eigenvalue λ .*

Proof. Define $x \in M$ and let $y \in N$ such that $x = h(y)$. Then

$$U_S \phi(x) = \phi(S(x)) = \lambda \phi(x)$$

$$\phi \circ (S \circ h)(y) = \lambda(\phi \circ h)(y)$$

$$\phi \circ (h \circ T)(y) = \lambda(\phi \circ h)(y)$$

$$U_T(\phi \circ h)(y) = \lambda(\phi \circ h)(y)$$

□

Some Math

How can we define similarity?

Procrustes Analysis over Vector Fields

A shape metric which compares the geometric similarity of two datasets after identifying an optimal orthogonal transformation* to align them.

$$\begin{aligned} d(\mathbf{A}_x, \mathbf{A}_y) &= \min_{\mathbf{C} \in O(n)} \|\mathbf{A}_x - \mathbf{C}\mathbf{A}_y\mathbf{C}^{-1}\|_F \\ &= \min_{\mathbf{C} \in O(n)} \arccos \frac{\mathbf{A}_x \cdot (\mathbf{C}\mathbf{A}_y\mathbf{C}^{-1})}{\|\mathbf{A}_x\|_F \|\mathbf{A}_y\|_F} \end{aligned}$$

- Zero: $d(A, B) = 0 \Leftrightarrow A = B$
- Symmetry: $d(A, B) = d(B, A)$
- Triangle Inequality: $d(A, C) \leq d(A, B) + d(B, C)$

- $d(\mathbf{A}, \mathbf{B})$ is a metric.

$g(A, B) = \|A - B\|_F$ is a metric (trivially true) & the similarity transform is an isometry

$$g(T(A), T(B)) = g(A, B)$$

$$g(T(A), T(B)) = \|CAC^{-1} - CBC^{-1}\|_F = \|C(A - B)C^{-1}\|_F$$

$$g(T(A), T(B)) = \text{Tr}([C(A - B)C^{-1}]^T [C(A - B)C^{-1}])$$

$$= \text{Tr}(C(A - B)^T C^{-1} C(A - B) C^{-1})$$

$$= \text{Tr}(C(A - B)^T (A - B) C^{-1}) = \text{Tr}(C^{-1} C(A - B)^T (A - B))$$

$$= \text{Tr}((A - B)^T (A - B)) = \|A - B\|_F = g(A, B)$$

→ If the transformations do not distort the original metric, then we can define a consistent distance between the transformed objects

* $O(n)$: The group of orthogonal matrices of dimension n . Refers to rotation if $\det(A) = 1$, and reflection if $\det(A) = -1$.

Some Math

How can we define similarity?

- **Euclidean distance to angular distance**

$$\begin{aligned}\|A - CBC^{-1}\|^2 &= \text{Tr}((A - CBC^{-1})^T(A - CBC^{-1})) \\ &= \text{Tr}(A^T A) - 2\text{Tr}(A^T CBC^{-1}) + \text{Tr}(B^T B) \\ &= \text{Tr}(A^T A) - \boxed{2 \langle A, CBC^{-1} \rangle} + \text{Tr}(B^T B) \\ &\quad \rightarrow \text{maximize} \\ \arccos \frac{\langle A, CBC^{-1} \rangle}{\|A\| \|B\|}\end{aligned}$$

→ Since arccos function monotonically decreases, maximizing the matrix inner product minimizes the angular distance.

Some Math

How can we measure similarity?

Step 1

$C \in SO(n)$ covers all the possible cases

We cannot continuously parametrize all of $O(n)$, as it is made up of two disconnected components $SO(n)$ where $\det(C) = 1$ and $O(n) \setminus SO(n)^*$.

Theorem 1. All elements of $O(n)/SO(n)$ are bijective maps between itself and $SO(n)$.

Proof. Let $D, P \in O(n)/SO(n)$. Then $PD \in SO(n)$. For this to be true, PD must have two properties: $\det(PD) = 1$, and $(PD)^T(PD) = I$. The first is easily shown: $\det(PD) = \det(P)\det(D) = -1 * -1 = 1$. The second is likewise shown easily: $(PD)^T(PD) = D^T P^T P D = D^T D = I$ by orthogonality. Both P and D have inverses P^T and D^T .

Conversely, let $D \in SO(n)$. Then we can show by the same logic that $PD \in O(n)/SO(n)$. $\square$

$$\Rightarrow \min_{Q \in O(n)} \|B - QAQ^T\|_F = \min \left\{ \begin{array}{l} \min_{C \in SO(n)} \|B - CAC^T\|_F, \\ \min_{C \in SO(n)} \|B - CPA P^T C^T\|_F \end{array} \right\}$$

* $SO(n)$ is a subgroup of $O(n)$ while $O(n) \setminus SO(n)$ is not, as the latter does not have the identity element.

Some Math

How can we measure similarity?

Step 2

Cayley Transform

For an arbitrary skew-matrix* $\mathbf{S}$, the matrix $\mathbf{C}$ calculated as below is orthogonal:

$$\mathbf{C} = (\mathbf{I} - \mathbf{S})(\mathbf{I} + \mathbf{S})^{-1}$$

- Cayley Transform restricts the operation to $SO(n)$:

$$\begin{aligned}\mathbf{C} &= (\mathbf{I} - \mathbf{S})(\mathbf{I} + \mathbf{S})^{-1} \Rightarrow \mathbf{C}^T = [(\mathbf{I} + \mathbf{S})^{-1}]^T (\mathbf{I} - \mathbf{S})^T = [(\mathbf{I} + \mathbf{S})^T]^{-1} (\mathbf{I} - \mathbf{S})^T \\ &\Rightarrow \mathbf{C}^T = (\mathbf{I} - \mathbf{S})^{-1} (\mathbf{I} + \mathbf{S}) \\ &\Rightarrow \mathbf{C}^T \mathbf{C} = (\mathbf{I} - \mathbf{S})^{-1} (\mathbf{I} + \mathbf{S})(\mathbf{I} + \mathbf{S})^{-1} (\mathbf{I} - \mathbf{S}) = \mathbf{I}\end{aligned}$$

$$\det(\mathbf{C}) = \det[(\mathbf{I} - \mathbf{S})(\mathbf{I} + \mathbf{S})^{-1}] = \frac{\det(\mathbf{I} - \mathbf{S})}{\det(\mathbf{I} + \mathbf{S})} = \frac{\det((\mathbf{I} - \mathbf{S})^T)}{\det(\mathbf{I} + \mathbf{S})} = 1$$

→ Re-parametrization allows computing the metric with gradient descent over $\mathbf{S}$ via Adam optimizer (e.g., `C = torch.linalg.solve(I+S)(I-S)`).

* **Skew-matrix:** Matrices that satisfy $X^T = -X$.

Results

DSA captures dynamical similarity and differences

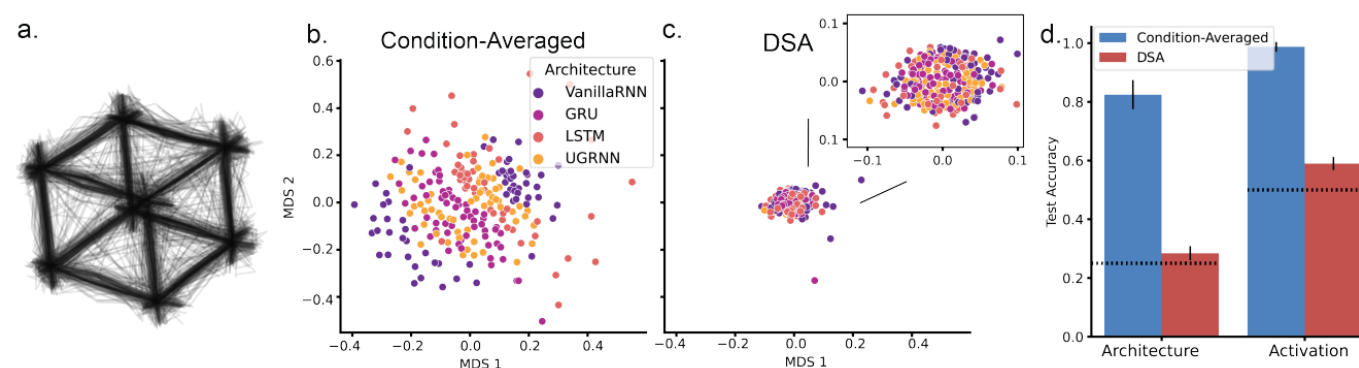


Figure 2: Same dynamics, different shape, only identified as similar with DSA.

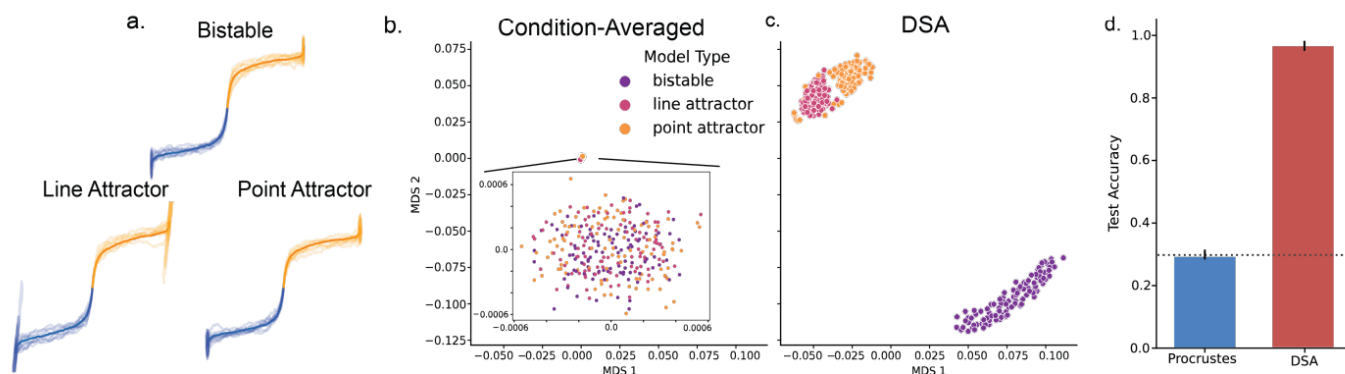


Figure 3: Different dynamics, same shape, only distinguished by DSA.

Results

DSA responds to topology changes, but does not to geometric deformations

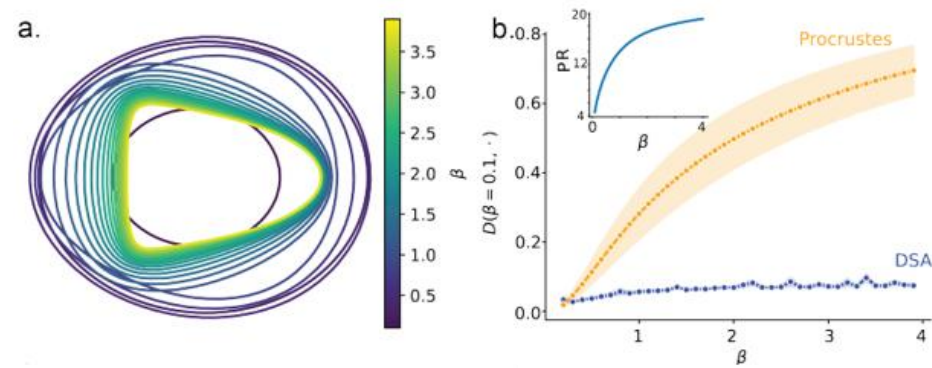


Figure 4a: DSA is invariant under geometric deformation of a ring attractor.

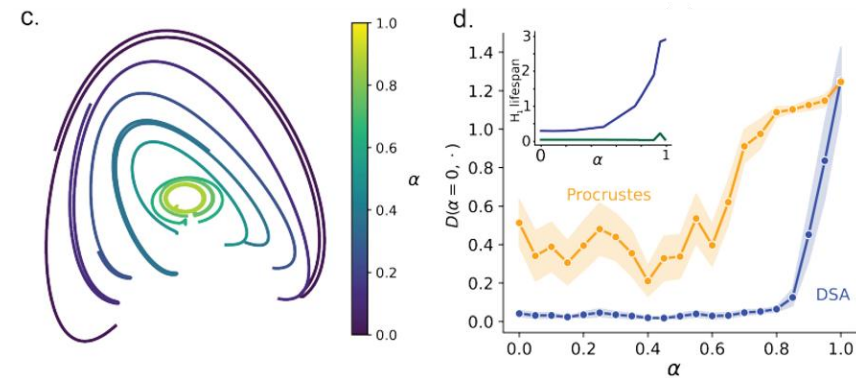


Figure 4b: DSA is sensitive to the transformation of a ring attractor into a line.

Results

DSA can disentangle learning rules in an unsupervised fashion

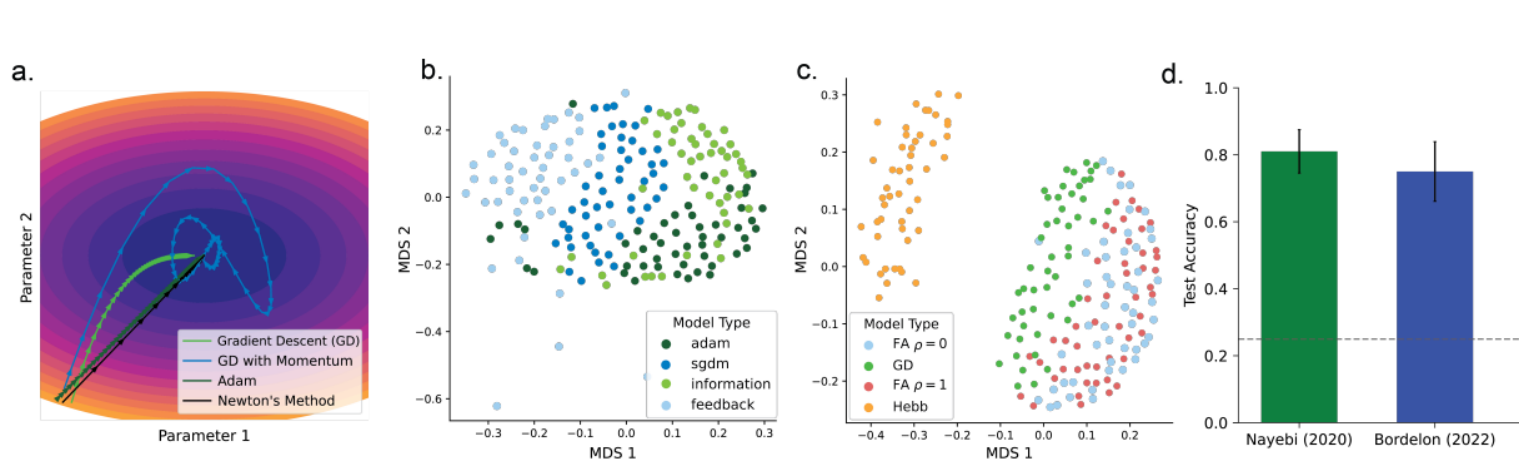


Figure 5: DSA unsupervisedly disentangles learning rules.

Thank you 😊

Appendix

References

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